

ALFAAS60241

## Glacial Acetic Acid

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 乙酸  
Glacial Acetic Acid

**Cat No. :** S60241  
**Synonyms** Ethanoic acid; Glacial acetic acid; Methanecarboxylic acid  
**CAS No** 64-19-7  
**Molecular Formula** C2 H4 O2

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
vinegar-like

#### Emergency Overview

Flammable liquid and vapor. Causes severe skin burns and eye damage. May be harmful if swallowed.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Flammable liquids.                | Category 3   |
| Acute Oral Toxicity               | Category 5   |
| Skin Corrosion/Irritation         | Category 1 A |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements



## Glacial Acetic Acid

## Signal Word

## Danger

## Hazard Statements

H226 - Flammable liquid and vapor

H314 - Causes severe skin burns and eye damage

H303 - May be harmful if swallowed

## Precautionary Statements

## Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P243 - Take action to prevent static discharges

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Vapors may cause flash fire or explosion. Flammable liquid.

## Health Hazards

Corrosive. Causes skin and eye burns. May be harmful if swallowed.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

## Other Hazards

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component   | CAS No  | Weight % |
|-------------|---------|----------|
| Acetic acid | 64-19-7 | >95      |

## SECTION 4. FIRST AID MEASURES

## General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

## Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

## Inhalation

## Glacial Acetic Acid

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

**Ingestion**

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Use personal protective equipment as required.

**Notes to Physician**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure. Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

## Glacial Acetic Acid

Corrosives area. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component   | China                                                   | Taiwan                                   | Thailand    | Hong Kong                                                                              |
|-------------|---------------------------------------------------------|------------------------------------------|-------------|----------------------------------------------------------------------------------------|
| Acetic acid | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> |

| Component   | ACGIH TLV                   | OSHA PEL                                                                                                 | NIOSH                                                                                                  | The United Kingdom                                                                     | European Union                                                                                                   |
|-------------|-----------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Acetic acid | TWA: 10 ppm<br>STEL: 15 ppm | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | IDLH: 50 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> | STEL: 37 mg/m <sup>3</sup><br>STEL: 15 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 25 mg/m <sup>3</sup> (8h)<br>TWA: 10 ppm (8h)<br>STEL: 50 mg/m <sup>3</sup> (15min)<br>STEL: 20 ppm (15min) |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Tight sealing safety goggles or Face protection shield Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Butyl rubber   | > 480 minutes     | 0.7 mm          | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

## Glacial Acetic Acid

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387                                       |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains.                                                                                                                                                                                                                                                                                       |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                               |                                          |
|------------------------------------------------|-----------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                                     |                                          |
| <b>Physical State</b>                          | Liquid                                        |                                          |
| <b>Odor</b>                                    | vinegar-like                                  |                                          |
| <b>Odor Threshold</b>                          | No data available                             |                                          |
| <b>pH</b>                                      | < 2.5                                         | 10 g/L aq.sol                            |
| <b>Melting Point/Range</b>                     | 16 - 16.5 °C / 60.8 - 61.7 °F                 |                                          |
| <b>Softening Point</b>                         | No data available                             |                                          |
| <b>Boiling Point/Range</b>                     | 117 - 118 °C / 242.6 - 244.4 °F               |                                          |
| <b>Flash Point</b>                             | 40 °C / 104 °F                                | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | 0.97 (Butyl Acetate = 1.0)                    |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 4 vol%<br><b>Upper</b> 19.9 vol% |                                          |
| <b>Vapor Pressure</b>                          | 1.52 kPa @ 20 °C                              |                                          |
| <b>Vapor Density</b>                           | 2.10                                          | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.048                                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                   |
| <b>Water Solubility</b>                        | Miscible                                      |                                          |
| <b>Solubility in other solvents</b>            | No information available                      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                |                                          |
| Acetic acid                                    | -0.2                                          |                                          |
| <b>Autoignition Temperature</b>                | 427 °C / 800.6 °F                             |                                          |
| <b>Decomposition Temperature</b>               | No data available                             |                                          |
| <b>Viscosity</b>                               | 1.53 mPa.s @ 25 °C                            |                                          |
| <b>Explosive Properties</b>                    |                                               | explosive air/vapour mixtures possible   |
| <b>Oxidizing Properties</b>                    | No information available                      |                                          |
| <b>Molecular Formula</b>                       | C2 H4 O2                                      |                                          |
| <b>Molecular Weight</b>                        | 60.05                                         |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

**SAFETY DATA SHEET****Glacial Acetic Acid**

|                                         |                                                                                                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                      |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.                              |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Strong bases. Metals.                                                                                     |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Thermal decomposition can lead to release of irritating gases and vapors. |

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component   | LD50 Oral          | LD50 Dermal | LC50 Inhalation       |
|-------------|--------------------|-------------|-----------------------|
| Acetic acid | 3310 mg/kg ( Rat ) | -           | > 40 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

**(e) germ cell mutagenicity;** No data available  
Not mutagenic in AMES Test

**(f) carcinogenicity;** No data available  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**SECTION 12. ECOLOGICAL INFORMATION**

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

**SAFETY DATA SHEET****Glacial Acetic Acid**

| Component   | Freshwater Fish                                                                          | Water Flea         | Freshwater Algae | Microtox                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------|--------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetic acid | Pimephales promelas:<br>LC50 = 88 mg/L/96h<br>Lepomis macrochirus:<br>LC50 = 75 mg/L/96h | EC50 = 95 mg/L/24h | -                | Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/15 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/25 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/5 min |

**Persistence and Degradability****Persistence****Degradation in sewage  
treatment plant**

Expected to be biodegradable

Miscible with water, Persistence is unlikely, based on information available.

Neutralization is normally necessary before waste water is discharged into water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Acetic acid | -0.2    | No data available             |

**Mobility in soil**

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information****Persistent Organic Pollutant****Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                         |                      |
|-------------------------|----------------------|
| UN-No                   | UN2789               |
| Proper Shipping Name    | ACETIC ACID, GLACIAL |
| Hazard Class            | 8                    |
| Subsidiary Hazard Class | 3                    |
| Packing Group           | II                   |

**IMDG/IMO**

|                         |                      |
|-------------------------|----------------------|
| UN-No                   | UN2789               |
| Proper Shipping Name    | ACETIC ACID, GLACIAL |
| Hazard Class            | 8                    |
| Subsidiary Hazard Class | 3                    |
| Packing Group           | II                   |

## Glacial Acetic Acid

IATA

UN-No UN2789  
Proper Shipping Name ACETIC ACID, GLACIAL  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group II

Special Precautions for User No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component   | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|------|
| Acetic acid | X                                                   | X                                       | X    | X     | 200-580-7 | X    | X   | X     | X    | X    | X    | X    |

**National Regulations****SECTION 16. OTHER INFORMATION**

Prepared By Health, Safety and Environmental Department  
Creation Date 05-May-2009  
Revision Date 13-May-2024  
Revision Summary New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%



## Glacial Acetic Acid

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**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**